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Twenty Dollars, No Difference

A Cluster-Randomised Trial of Booking Deposits Across 38 Hair Salons, Against the Before-and-After Figure That Steered the Rollout

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Abstract. Appointment deposits are widely adopted by small service businesses on the reasoning that a client who stands to lose money will turn up, and they are almost never evaluated against a concurrent control. We report a cluster-randomised trial of a $20 booking deposit across the 38 salons of one metropolitan hairdressing chain, with the deposit taken at booking, redeemable against the service, and forfeited on non-attendance. Salons were allocated 19 to the deposit and 19 to booking on trust, stratified by chair count and trading pattern, with the analysis specified before allocation. Across 61,430 booked appointments in the twenty-one weeks of concurrent operation, the no-show rate was 7.6% in the deposit arm and 7.9% in the control arm, an intention-to-treat difference of −0.31 percentage points (95% CI −1.18 to 0.56), which falls inside the pre-specified equivalence margin of ±1.5 points. Over the same period the chain's own before-and-after comparison within the deposit salons returned a fall of 3.2 points, a figure that appears in full in the control salons as well. The deposit was extended to the whole portfolio in week 22 on the strength of the before-and-after figure, four weeks before the trial reported. Our results suggest the deposit may be doing nothing that the calendar was not already doing, though the direction of the estimate is consistent across every specification we examined.

Introduction

A booking deposit is one of the few pieces of behavioural policy a small business can implement in an afternoon. The chain we study did so through a setting in its booking software: a $20 charge at the point of booking, held against the client's card, redeemed against the service on arrival, forfeited if the client neither attends nor cancels with a day's notice. Nobody has to be persuaded, trained, or reminded. The mechanism is intuitive, its rationale is loss aversion [1], and the number it targets — the share of booked appointments nobody turns up for — is already in the weekly report.

That combination is precisely what makes deposits hard to evaluate and easy to adopt. The intuition is strong enough that a business rarely holds any salons back, and once every site has the deposit, the only comparison available is the site with itself before it. That comparison is known to be unreliable in exactly the setting where deposits are introduced — after a bad quarter, at a seasonal peak, when the number has nowhere to go but down [2], [3] — and its standard errors are optimistic besides [4].

There is also reason to doubt the mechanism. Penalties introduced to deter a behaviour sometimes license it instead, by converting an obligation into a priced option [5], the conditions under which financial incentives change behaviour are narrower than their popularity suggests [6], and behavioural interventions evaluated at scale return effects substantially smaller than the published literature predicts [7], [8].

We contribute the following, each qualified as the design allows:

- A cluster-randomised trial of a booking deposit in a commercial service setting, which finds no difference in no-show rate large enough to matter, and is powered to say so.
- A direct comparison of the trial estimate with the before-and-after estimate the chain computed from the same data, suggesting the whole of the latter is attributable to the period rather than the deposit.
- A record of the decision the chain took while the trial was running, which we report as the outcome it is rather than as a limitation of the study.

Background

Missed appointments are a mature research object in health services, where the reminder is the intervention of record: telephone and text reminders raise attendance at booked appointments across a wide range of settings [9], [10]. Deposits and penalties are far less studied, and almost never with a concurrent control, which is the gap this paper occupies.

The methodological ground is well laid. Cluster-randomised designs and their sample-size requirements are standard [11], [12], their reporting is codified [13], and the analysis convention — intention to treat — is long settled [14]. Where a trial is expected to return no difference, the appropriate machinery is equivalence testing rather than a failure to reject [15], since a non-significant result and a demonstrated absence are different claims [16], and retrospective power calculations are not a substitute for either [17], [18]. What a randomised estimate does and does not license has been argued at length [19], as has the choice between it and the observational alternatives an organisation usually has to hand [20], [21].

Prior work at this institution has repeatedly found the same gap between a booking system's numbers and the behaviour they stand for. A suburban gym chain's booking-reliability score raised booking-fill rates at popular classes from 93.6% to 99.3% while door-scanner attendance held flat, with cancellations migrating to the six-minute band just outside the score's penalty cutoff [22]; and a market's undifferentiated no-show count was found to predict recurrence far less accurately than a four-category taxonomy of the reasons behind it [23]. Whether organisations use the evidence available to them, rather than the evidence already formatted for their reporting cycle, is a question older than any of this [24].

Method

Setting and allocation. The chain operates 38 salons across one metropolitan area, ranging from two to eleven chairs, all on a single booking platform with a common no-show definition. Salons were allocated 19 to the deposit arm and 19 to continue booking on trust, stratified by chair count (2, 3–4, 5–6, 7+) and by whether the salon trades on Sundays. Allocation was performed by an analyst outside the study team from a fixed seed, after the analysis plan had been lodged with the School's Evaluation of Evaluation programme. Clients were not blinded, which is a property of the intervention rather than a design choice; salon staff were briefed by the chain in a single joint session.

Intervention. From the launch date, salons in the deposit arm required a \$20 deposit at the point of booking, taken by the booking platform for online bookings and by card at the counter for bookings made by telephone or in person. The deposit was redeemed against the service on attendance and forfeited if the client did not attend, or cancelled less than 24 hours ahead. Control salons continued to book without a deposit.

Outcomes. The primary outcome, specified in advance, was the no-show rate: booked appointments that were neither attended nor cancelled more than 24 hours before their start time, as a share of all booked appointments. Secondary outcomes were bookings per chair-hour, the share of clients rebooking at the chair before leaving, and the share of cancellations lodged 24 to 72 hours ahead — the band into which a deposit would be expected to push a client who had decided not to attend.

Analysis. Appointment i at salon j was modelled with a mixed-effects logistic regression carrying a salon random intercept and the stratification fixed effects $\gamma_{s(j)}$:

$$\operatorname{logit} \Pr(y_{ij} = 1) = \beta_0 + \beta_1 D_j + \gamma_{s(j)} + u_j,$$

with D_j the deposit indicator and $u_j \sim N(0, \sigma_u^2)$. The intracluster correlation was 0.021, close to the 0.02 assumed at design. Equivalence was assessed against a margin of ±1.5 percentage points, fixed in the protocol as the smallest difference the chain's own operations team described as worth having. The contrast the chain computes, and the contrast the trial computes, are distinct estimands over the same records:

$$\hat{\delta}_{\text{pre/post}} = \overline{y}^{D}_{\text{post}} - \overline{y}^{D}_{\text{pre}}, \quad \hat{\delta}_{\text{ITT}} = \overline{y}^{D}_{\text{post}} - \overline{y}^{C}_{\text{post}}.$$

Window. The trial was designed to run 26 weeks. In week 18 the chain approved extension of the deposit to the whole portfolio, and control salons began taking deposits in week 22, so the primary analysis covers the 21 weeks of concurrent operation. The full window is retained as a sensitivity analysis.

Results

Across the 21 weeks of concurrent operation the two arms recorded 61,430 booked appointments, 30,712 in the deposit arm and 30,718 in the control arm. The no-show rate was 7.6% under the deposit and 7.9% without it, an intention-to-treat difference of −0.31 percentage points (95% CI −1.18 to 0.56). The mixed-effects odds ratio is 0.96 (95% CI 0.87 to 1.06). The confidence interval lies inside the ±1.5-point equivalence margin in both directions (TOST p = 0.004), so the trial supports equivalence rather than merely failing to find a difference.

The weekly series shows why the intuition survives (Figure 1). Both arms enter the trial at roughly 11% and leave it below 7.5%, a descent that begins before the deposit exists and continues at the same slope through every week of concurrent operation. The signed band between the two series changes direction eleven times in 21 weeks and never exceeds 0.7 points.

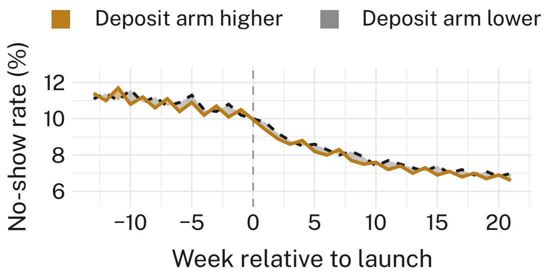

Figure 1: Weekly no-show rate in each arm. The band between them crosses zero eleven times and never exceeds 0.7 points; the shared fall from 11% to below 7.5% is present in salons that never took a deposit.

The before-and-after contrast returns a much larger number. Within the deposit salons, the no-show rate fell from 10.8% in the thirteen weeks before launch to 7.6% after it, a reduction of 3.2 percentage points, or 29.6% in relative terms. This is the figure the chain reported. The same contrast computed on the control salons, which took no deposits at any point in the window, returns 10.9% falling to 7.9% — a reduction of 3.0 points. The pattern holds in every stratum (Figure 2): the four size bands move together, and the arms within each band are separated by less than the bands are separated from each other.

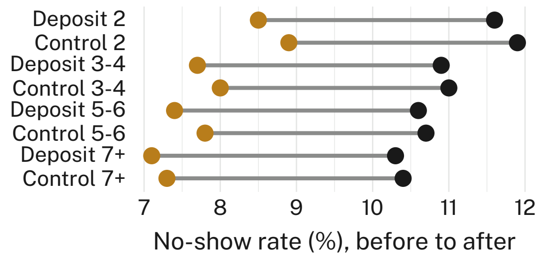

Figure 2: No-show rate before and after launch in each size band of each arm. Every band falls by between 2.9 and 3.2 points, whether or not it took deposits; the largest deposit-to-control gap after launch is 0.5 points, in the smallest salons.

Secondary outcomes are reported in Table 1. Bookings per chair-hour fell 2.1% in the deposit arm, and the share of clients rebooking at the chair before leaving fell 2.9 points, both differences excluding zero. The cancellation-band outcome moved in the direction a deposit would predict but not distinguishably so.

Outcome	Deposit	Control	Difference (95% CI)
No-show rate	7.6%	7.9%	−0.31 (−1.18, 0.56)
Bookings/ chair-hour	1.87	1.91	−2.1% (−3.6, −0.5)
Rebooked at chair	41.2%	44.1%	−2.9 (−4.6, −1.2)
Cancelled 24–72 h	6.4%	5.8%	0.6 (−0.2, 1.4)

Table 1: Primary and secondary outcomes over the 21 weeks of concurrent operation. The deposit is associated with fewer bookings and fewer rebookings, and with no change in the outcome it was introduced to change.

Seven pre-specified analysis variants were fitted (Figure 3). Estimates range from −0.42 to +0.22 percentage points; every interval contains zero and every point estimate falls well inside the equivalence margin. Per-protocol delivery moves the estimate by 0.11 points, and excluding the two salons that waived the deposit for regular clients moves it by 0.07. We retain the intention-to-treat specification as primary for completeness. Salon-level estimates are similarly unhelpful to the mechanism: the median salon difference is −0.2 points (IQR −1.9 to 1.4), and 4 of 38 salons reach individual significance at the 5% level, two in each direction, against 1.9 expected by chance.

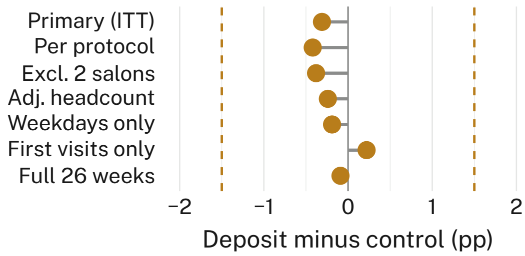

Figure 3: The primary estimate under seven pre-specified analysis variants. The widest spread between any two variants is 0.64 percentage points, less than half the equivalence margin (dashed).

Discussion

The deposit did not change the no-show rate, and the trial is precise enough to say so rather than merely to fail to detect a change. What it did change, on our secondary outcomes, is the volume of bookings and the rate at which clients rebook before leaving the salon — both small, both in the unfavourable direction, and neither reported in the operations pack, which carries the no-show rate alone.

The more interesting finding is the arithmetic of the number that was reported. A fall of 3.2 points inside the deposit salons is real, correctly computed, and entirely present in salons that never took a deposit. The chain's quarterly pack tabled it in week 17 as a 29.6% reduction; the extension was approved in week 18 and implemented from week 22. We observe consistent behaviour across all four size bands and every analysis variant, which suggests a property of the intervention rather than of any particular salon, though the smallest salons warrant attention.

We record this without criticism of the chain, which computed the only estimate its own reporting produces on a cycle short enough to act on. The trial's estimate arrived four weeks later, and answered a question nobody was asking.

Limitations

Our analysis covers 21 of the 26 planned weeks, though the estimate is stable from week nine onward and the truncation cost precision rather than direction. The trial ran in a single metropolitan chain, whose common booking platform is what made a clean no-show definition possible across 38 sites — an unrepresentative feature that is also the reason the outcome can be trusted. We did not survey clients about the deposit, having judged that a client's account of why they missed an appointment is a weaker record than whether they did.

Conclusion

A $20 booking deposit, allocated at random across 38 salons, changed the no-show rate by −0.31 percentage points (95% CI −1.18 to 0.56), within a pre-specified equivalence margin, while reducing bookings and rebookings by small but measurable amounts. The before-and-after figure computed on the same records returned a reduction ten times larger, all of which is present in the untreated arm, and it was that figure the portfolio decision cited. Future work will ask whether a deposit's effect on booking volume is worth measuring on its own, and will extend the concurrent-control design to the other settings the chain's platform reports on.

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